Md. Mosfikur Rahman

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Research Interests

Machine learning for large-scale distributed systems: anomaly detection, failure prediction and resource management in production microservice infrastructure, and in systems security.

Education

2018 – 2021 **B.Sc. in Computer Science and Engineering**
Daffodil International University, Dhaka, Bangladesh
Four-year programme. CGPA 3.83 / 4.00

2021 **Exchange semester** (bilateral exchange with Daffodil International University)
Adam Mickiewicz University, Poznań, Poland

Technical Skills

Modelling Machine learning: supervised and deep learning, CNNs, anomaly and intrusion detection, applied NLP, data preprocessing and visualisation

Systems Distributed systems, microservice architecture, multi-tenancy, REST, GraphQL, gRPC, Server-Sent Events, observability and distributed tracing, Docker, Kubernetes, Azure, AWS

Languages Python, C#, Java, TypeScript, SQL

Data SQL Server, PostgreSQL, Oracle, MySQL, MongoDB, DynamoDB, Redis, Elasticsearch

Certifications IBM Data Science Professional Certificate (Coursera)

Professional Experience

2024 – Present **Engineering Team Lead** (Software Engineer L-II), Akij iBOS Ltd., Dhaka
Architected a multi-tenant travel platform of eight .NET 9 microservices, from an empty repository to three live deployments serving 742 B2B agency accounts. Designed the flight-search aggregation layer, fanning out over gRPC to eleven supplier and GDS connectors (Sabre, Amadeus, Travelport) over Server-Sent Events: 1,000+ fares in under ten seconds, the first in approximately four. Modelled pricing and discount logic as tenant configuration, moving 96 live rules out of the deploy path. Built the observability layer capturing every inter-service call, now the primary production diagnosis tool. Led a team of ten.

2023 – 2024 **Software Engineer**, REVE Systems Ltd., Dhaka
Delivered the Legal Case and Utilization Declaration modules of the National Board of Revenue's Customs Bond Management System in Java, Spring Boot and Oracle, replacing a manual compliance process for garment exporters. Improved system throughput by approximately 25% through query tuning and data-access optimisation.

2022 – 2023 **Software Engineer** (Backend), BJIT Group, Dhaka
Built a GraphQL Backend-for-Frontend for Rakuten e-commerce modules, cutting nested-query latency through resolver batching and schema stitching.

Publications

Ten peer-reviewed publications, three as first author; author name in bold. **185 citations, h-index 4, i10-index 4** (*Google Scholar, September 2026*).

Journal Articles

[J3] **Rahman, Md. Mosfikur**, Saifuzzaman, M., Ahmed, A., Mahin, M. F., & Shetu, S. F. (2021). Impact of COVID-19 on mental health: A quantitative analysis of anxiety and depression based on regular life and internet use. *Current Research in Behavioral Sciences*, 2, 100037. Elsevier. doi:10.1016/j.crbeha.2021.100037
First author.

[J2] Shetu, S. F., **Rahman, Md. Mosfikur**, Ahmed, A., Mahin, M. F., Akib, M. A. U., & Saifuzzaman, M. (2021). Impactful e-learning framework: A new hybrid form of education. *Current Research in Behavioral Sciences*, 2, 100038. Elsevier. doi:10.1016/j.crbeha.2021.100038

[J1] Saifuzzaman, M., **Rahman, Md. Mosfikur**, Shetu, S. F., & Moon, N. N. (2021). COVID-19 and Bangladesh: Situation report, comparative analysis, and case study. *Current Research in Behavioral Sciences*, 2, 100034. Elsevier. doi:10.1016/j.crbeha.2021.100034

Conference Papers and Book Chapters

[C7] Johora, F. T., Mahfuja, I. B., Masuqur Rahman, A. N. M., **Rahman, Md. Mosfikur**, & Rahman, M. S. (2024). Machine Learning-Based Prediction of COVID-19: A Robust Approach for Early Diagnosis and Treatment. In *Proceedings of the Fifth International Conference on Trends in Computational and Cognitive Engineering* (pp. 205–215). Springer Nature Singapore. doi:10.1007/978-981-97-1923-5_16

[C6] Johora, F. T., Alamin-Ul-Islam, Yesmin, F., & **Rahman, Md. Mosfikur** (2023). A New Chaotic-Based Analysis of Data Encryption and Decryption. In *Advances in Data Science and Artificial Intelligence* (pp. 455–468). Springer International Publishing. doi:10.1007/978-3-031-16178-0_32

[C5] Islam, T., **Rahman, Md. Mosfikur**, Jabiullah, M. I., & Saifuzzaman, M. (2023). Cyber Security Intruder Detection Using Deep Learning Approach. In *Information Systems and Management Science* (pp. 518–530). Springer International Publishing. doi:10.1007/978-3-031-13150-9_42

[C4] Yesmin, F., **Rahman, Md. Mosfikur**, Saifuzzaman, M., & Moon, N. N. (2022). Pandemic Effect on Education System Among University Students. In *Progresses in Artificial Intelligence & Robotics: Algorithms & Applications* (pp. 157–168). Springer International Publishing. doi:10.1007/978-3-030-98531-8_16

[C3] **Rahman, Md. Mosfikur**, Foysal, M. R., Moon, N. N., & Nur, F. N. (2021). TraFoo: An Android Application for Food Delivery in Train. In *2021 International Conference on Smart Generation Computing, Communication and Networking (SMART GENCON)* (pp. 1–7). IEEE. doi:10.1109/SMARTGENCON51891.2021.9645900
First author.

[C2] Moon, N. N., Sharmin, S., Hossain, R. A., Jahan, I., Nur, F. N., & **Rahman, Md. Mosfikur** (2021). Deep Learning Model for Detecting and Diagnosing Plant Disease. In *2021 International Conference on Smart Generation Computing, Communication and Networking (SMART GENCON)* (pp. 1–8). IEEE. doi:10.1109/SMARTGENCON51891.2021.9645857

[C1] **Rahman, Md. Mosfikur**, Abul Kashem, M., Mohiuddin, M., Hossain, M. A., & Nessa Moon, N. (2020). Future City of Bangladesh: IoT Based Autonomous Smart Sewerage and Hazard Condition Sharing System. In *2020 IEEE International Women in Engineering (WIE) Conference on Electrical and Computer Engineering (WIECON-ECE)* (pp. 126–130). IEEE. doi:10.1109/WIECON-ECE52138.2020.9397950
First author. ***Best Paper Award.***

Awards and Honours

2021 **Winner, Data Science Hackathon**
Data Science Summit, Daffodil Smart Lab — national university-level competition.

2020 **IEEE Best Paper Award**
IEEE WIECON-ECE 2020, for [C1].

Peer Review

Reviewed 78 manuscripts: 49 for international journals and 29 for conferences.

Journals
ISA Transactions, Elsevier Q1 (7)
Journal of King Saud University – Computer and Information Sciences (12)
Natural Language Processing Journal (1)
Current Research in Behavioral Sciences, Elsevier (29)

Conferences
4th International Conference on Deep Learning, Artificial Intelligence and Robotics, 2022 (15)
International Conference on Information Systems and Management Science, 2022 (9)
International Conference on Communication and Information Systems, 2022 (5)

Invited Talks and Presentations

2024 **Conference Speaker**, 10th IEEE International Women in Engineering Conference on Electrical and Computer Engineering (WIECON-ECE 2024).

2021 **Guest Speaker**, "How to Get Started: The Undergraduate Research Journey." Computer & Programming Club, Daffodil International University.

Mentoring and Academic Leadership

2024 – Present **Engineering Team Lead**, Akij iBOS Ltd.
Established the code review and pair-programming practice the team works to.

2021 **Vice President, Research & Career Wing**, Computer & Programming Club, Daffodil International University
Led research workshops and mentorship programmes for undergraduates; advised peers on publishing, academic writing, and conference presentation.